

Sampling and Statistics

- In a typical statistical problem, we have a random variable X of interest but its pdf $f(x)$ or pmf $p(x)$ is not known. But the form of $f(x)$ or $p(x)$ is known down to a parameter θ (maybe a vector).
 - X has an exponential distribution, $\exp(\theta)$.
 - X has $b(n, p)$ distribution, where n is known but p is unknown.
 - X has Gamma distribution $\Gamma(\alpha, \beta)$, where both α and β are unknown.
- We want to estimate θ from a random sample.

- A random sample of size n : $X_1, \dots, X_n$ are independently and identically distributed (i.i.d.). $X_1, \dots, X_n \stackrel{iid}{\sim} f(x; \theta)$
- Statistic: $T = T(X_1, \dots, X_n)$. e.g. $\bar{X} = \frac{1}{n}(X_1 + \dots + X_n)$
- Estimator: T is called a point estimator of θ .
- Unbiased estimator: $\mathbb{E}(T) = \theta$. e.g. $\mathbb{E}(\bar{X}) = \mu$
- We will discuss maximum likelihood estimation (MLE).

An Motivation Example

- An urn contains m balls labeled from 1 to m . The experiment is to choose a ball at random and record the number. Let X denote the number. Then the distribution of X is given by

$$\mathbb{P}(X = x) = \frac{1}{m}, x = 1, \dots, m.$$

We do not know m and want to estimate m . So, $\theta = m$. We take a sample of n balls denoted by $\mathbf{X} = \{X_1, \dots, X_m\}$.

- **Sampling with Replacement:** Here a ball is selected at random, its number is recorded, the ball is replaced in the urn, the balls are then remixed.
- **Sampling without Replacement:** Here a ball is selected at random. If the balls are selected one-at-a-time, they are not replaced after each draw.
- When m is much larger than n , the sampling schemes are practically the same.

- To estimate m we draw $X_1, \dots, X_n$ with replacement. The distribution of X_i is $\mathbb{P}(X_i = x) = 1/m$ for $x = 1, \dots, m$. An intuitive estimator of m is the statistics $T = \max\{X_1, \dots, X_n\}$. How far is T from m ? We will prove that T is consistent estimator of m , $T \xrightarrow{P} m$ as $n \rightarrow \infty$.
- In the above problem, $\mathbb{E}(X) = (m+1)/2$. Hence, $\mathbb{E}(2\bar{X} - 1) = m$. Perhaps, $2\bar{X} - 1$ is also a good estimator of m as it is unbiased. We will show that T is a better estimator in this case.